

Date: Monday 20, 11/2017

Name:

Q1) choose the correct answer (12 marks)

1	2	3	4	5	6	7	8
---	---	---	---	---	---	---	---

- Which of the following number equivalent to $(276)_{10}$?
a) $(276)_H$ b) $(BE)_H$ c) 010111101 d) $(192)_4$
- Which of the following number equivalent to $(1ABF)_H$?
a) 110101011111 b) 101010111111 c) 110101011111 d) 110101001111
- The maximum value can be represented from 9 bit number is
a) 511 b) 512 c) 1024 d) 1023
- Which of the following function is equivalent to $F(A,B,C) = \sum m(1,3,4,5)$
A. $F(A,B,C) = A'B'C' + A'BC + AB'C' + ABC$
B. $F(A,B,C) = A'B'C + A'BC + AB'C' + AB'C$
C. $F(A,B,C) = A'B'C' + AB'C + A'BC + ABC$
D. $F(A,B,C) = A'B'C' + AB'C + A'BC' + ABC'$
- Which of the following number is equivalent to $(328)_{10}$
a) 101001000 b) 101001100 c) 101011100 d) 101101000
- The 10^{th} complement of 2643 number is
a) 7356 b) 7357 c) 7355 d) 7358
- Which of the following function will be get from the k-map
a) $F = \sum(2, 5, 11, 12)$ b) $F = \sum(2, 5, 8, 15)$ c) $F = \sum(3, 5, 10, 12)$
d) $F = \sum(3, 5, 8, 14)$
- We cant get the full adder from
a) 2 half adder and 2 OR gate
b) one half adder and 2 AND gate
c) 2 half adder and 1 OR gate
d) 2 half adder and 1 AND gate

CD \ AB	00	01	11	10
00				1
01		1		
11	1			
10			1	

Q2) fill in gap (6 marks)

- Convert decimal 59.72 to BCD
- To implement XNOR gate using minimum number NANDs. How many NANDs we need?
- An equivalent representation for Boolean $AB+1$ is:
- An equivalent representation of $F(A,B,C)=(AB+C).C'$ is:
- Convert $(1100101000110101)_2$ to hexadecimal
- If $F(A,B,C,D)=\sum m(0,1,8,9,11)$; the $F' = \prod M($)

Q3) what is the simplified function of the following K-map using product of sum (4 marks)

CD \ AB	CD			
	00	01	11	10
00	1	1		X
01	x	x		
11				
10				1

Q4) Design the circuit needed to implement the following where x, y and z are 3bits numbers, using block adder, comparator and basic gate? 8 marks

If $(x >= 2)$

$F = (x+y) * z;$

Else

$F = (x-y) * z;$

Q5) Find $(-112)_{10} - (78)_{10} : 4\text{marks}$

a. Using 1's compliment

b. Using 2's compliment

Q6) Design a combinational circuit that takes a 3 bit number $X = X_3X_2X_1$, and one output $F = 1$ when the input is dividable by 3 with remainder = 0 OR when the XNOR of the 3 number is 1 ; otherwise $F=0$. (6 marks)

Q1:-

$$\textcircled{1} (276)_{\text{octal}} \Rightarrow (\underline{010} \quad \underline{111} \quad \underline{110}) \Rightarrow \begin{matrix} 1011 & 1110 \\ \uparrow & \uparrow \\ (B & E) \end{matrix}_{16}$$

كل رقم 3 بت كل 3 بت رقم واحد hexa-deci

(b)

$$\textcircled{2} (1ABF)_H \Rightarrow (001 \ 1010 \ 1011 \ 1111)_2$$

كل رقم 4 بت

(c)

$$\begin{aligned} (15) F &\rightarrow 1111 \\ (11) B &\rightarrow 1011 \\ (10) A &\rightarrow 1010 \\ 1 &\rightarrow 0001 \end{aligned}$$

$$\textcircled{3} 2^9 - 1 = 511$$

rule

number of bits
2 - 1

وهكذا بالتقريب

$$\textcircled{4} F(A, B, C) = \sum_m (1, 3, 4, 5) = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C$$

(b)

$$\textcircled{5} (328)_{10} \Rightarrow (101001000)$$

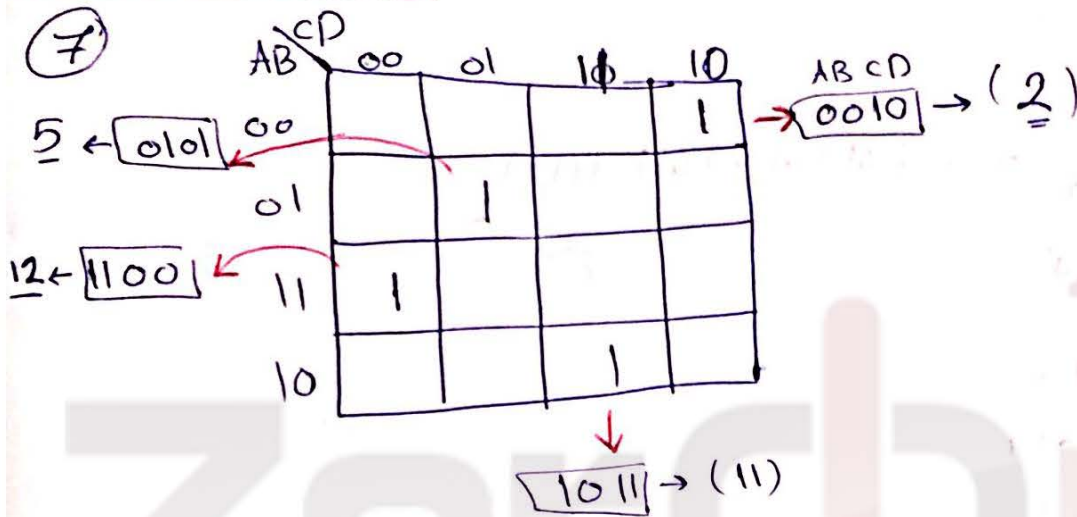
أو قسمة المتكررة

256	128	64	32	16	8	4	2	1
1	0	1	0	0	1	0	0	0

11

⑥ 2643
 $9's \text{ comp} \rightarrow 7356$
 $10^{th} \text{ comp} \rightarrow \begin{array}{r} 7356 \\ +1 \\ \hline 7357 \end{array}$

Ⓐ



$F = \sum (2, 5, 11, 12)$ Ⓐ

⑧ we can get the full adder from 3- (عدلو ابالامكان)

- 2 half adder and 1 OR gate.

Q2: Ⓐ $(59 \cdot 72)_{10} \Rightarrow (0101 \ 1001 \cdot 0111 \ 0010)_{BCD}$
 BCD $\frac{1}{2}$ بت

2

② XNOR by Nand Gates (أقل عدد) ؟!

* أقل عدد دارة لتتيل XOR من خلال Nand هو 4 دارة (أقل دارة)
 دارة بنفي المخرج من خلال Nand $[\neg]$ → بجای، الهراقة 6
 وبالتالي أقل عدد هو 5

③ $AB + 1 = 1$

④ $(AB + C) \cdot \bar{C} \Rightarrow AB\bar{C} + \cancel{C\bar{C}} = AB\bar{C}$
 ← بوزع الهمزة على الجميع

⑤ $(\underline{1100} \underline{1010} \underline{0011} \underline{0101})_2 \Rightarrow (CA35)_{16}$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 C A 3 5

⑥ $F(A, B, C, D) = \sum m(0, 1, 8, 9, 11)$

المطلوب $\boxed{\bar{F}} = \prod M(0, 1, 8, 9, 11)$ ← نفهم

← تداخل

« على استيادها

لهم نفع في winter ومن حادثة " .

0 → مزنیف
1 → Comp

* (وہمکن اعل sop و امولہاار pos .

Q4

(اگر کوئی یاد ہے)

نہیں ترتیب اور انکار

* لو کان آئل بھل $(x-y)*z$ و else بھل $(x+y)*z$

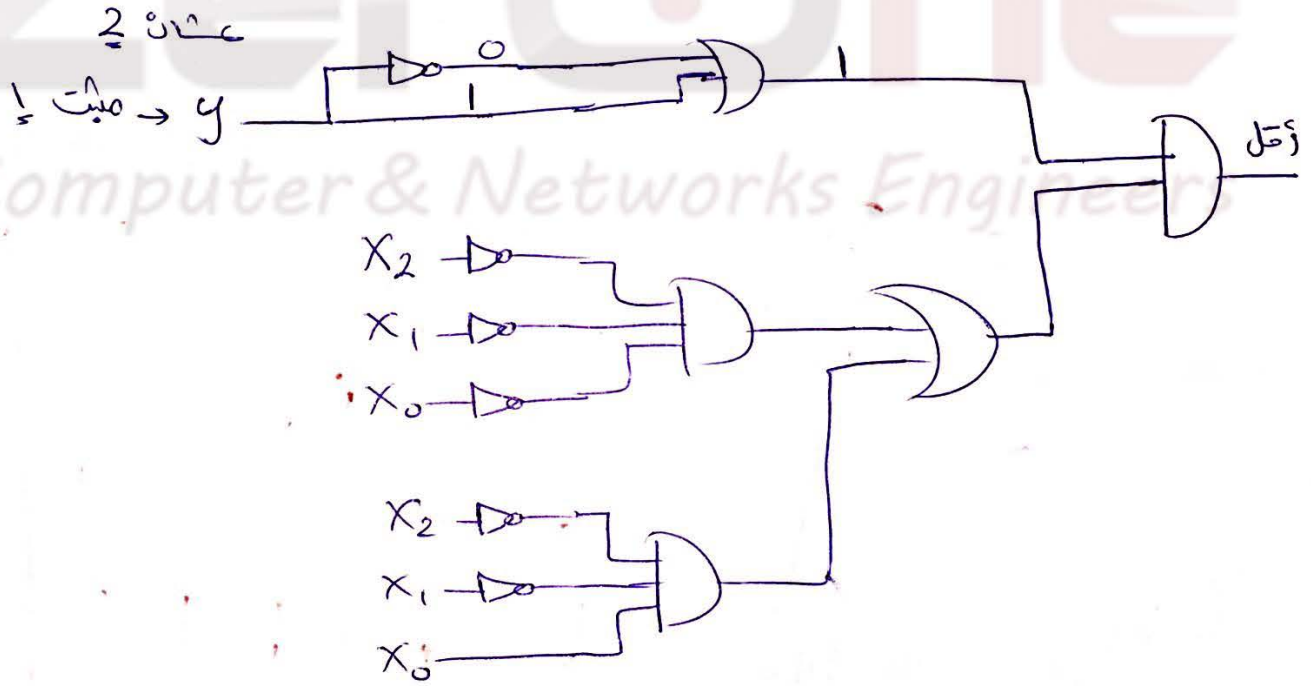
$X \rightarrow 3 \text{ bit}$

x_2	x_1	x_0	$\frac{z}{\leftarrow}$	آئل
0	0	0	1	1
0	0	1	1	1
0	1	0	1	0
1	0	0	1	0

* صی آئل ؟!

بی جالیہ ماریون
one & zero

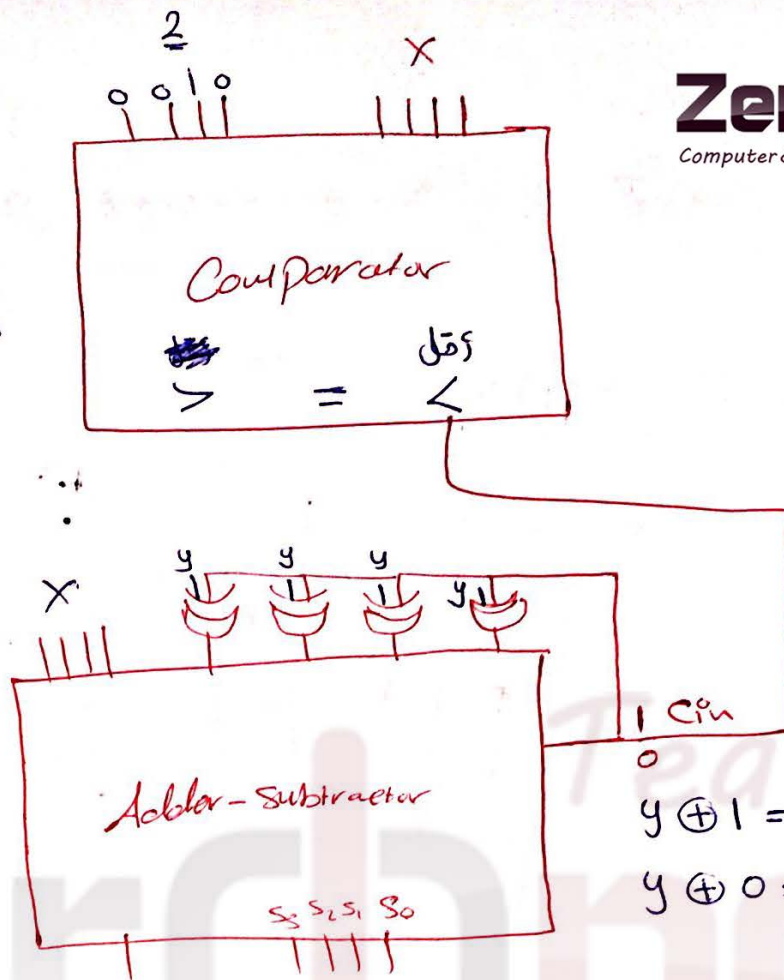
$$F = \bar{x}_2 \bar{x}_1 \bar{x}_0 + \bar{x}_2 \bar{x}_1 x_0$$



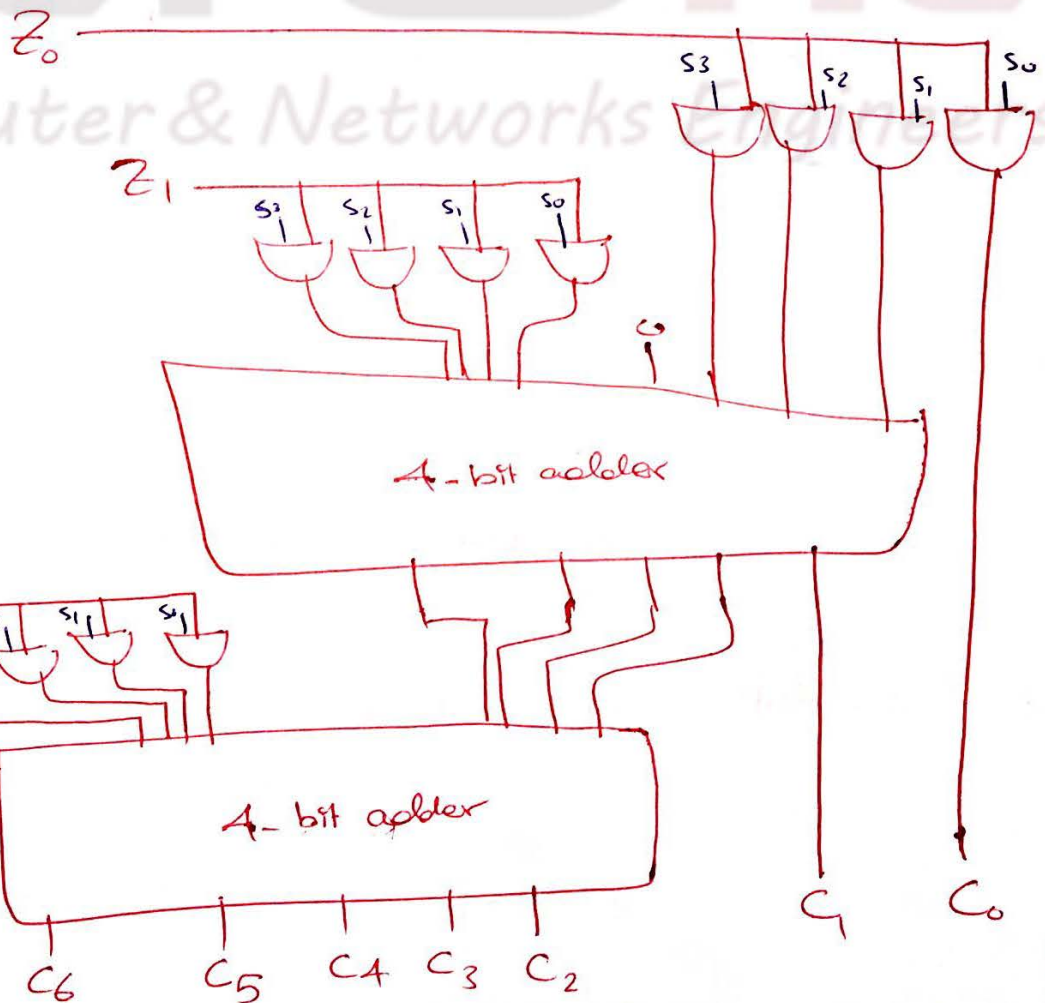
↑ حلا الخرج اے آئی کے باسی، بی صیٹ بدل کل الخوات اے

IC اور Comparator حلا هن

هلا انا اقل بفرع والعكس
مجمع 6 فبستدوم دائرة
[Adder-subtractor]



$Z = 3 \text{ bits}$
 3×4



$\boxed{F} \rightarrow$

Q5:-

$$(-112)_{10} - (78)_{10} = ? \quad \text{المخروم -190}$$

@

$$(112)_{10} \Rightarrow (01110000)_2 \Rightarrow \text{First Comp } (10001111)$$

64	32	16	8	4	2	1
1	1	1	0	0	0	0

↑
زيت
bit
عكس
مايعير
over flow

$$(78)_{10} \Rightarrow (01001110)_2 \Rightarrow \text{First Comp } (10110001)$$

64	32	16	8	4	2	1
1	0	0	1	1	1	0

↑
زيت
bit
عكس
مايعير
over flow

-112	1111111	10001111
-78	10110001	+
	01000000	1+
	01000001	

✓ لـ 1 ب

$$128 + 64 + 32 + 16 + 2 + 1 = 190 \checkmark$$

لو زيت
1st comp
لكن كذا

(b)

$$(112)_{10} \rightarrow (01110000)_2 \rightarrow \text{1st comp } (10001111)$$

$$(78)_{10} \rightarrow (01001110)_2 \rightarrow \text{1st comp } (10110001)$$

1+

(10010000)

(10110010)

7

-112

10010000

-78

10110010

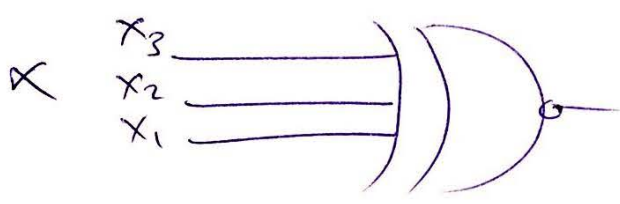
X 01000010
 ← سب
 کو بی
 دیکھ جائے
 1st comp
 ونیز واحد

10111101
 1+
 10111110
 ↓
 190 ✓
 5

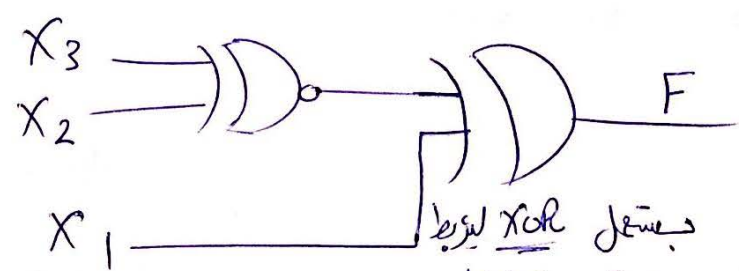
F

Q6 :-

X ₃	X ₂	X ₁	dividable by 3	XNOR	[/3] OR XNOR
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	1	1
1	1	1	0	0	0



کو آئیے ان
 XNOR
 دیکھ جائے



بستل
 XNOR
 دیکھ جائے
 T to T

8